WORKED SOLUTION

STEP II Specimen paper, Question 11

Mechanics • Projectiles and Collisions

STEP II Specimen paper, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

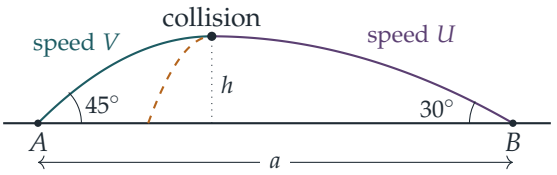
step.maths.org/questions/spec-s2-q11

The key idea

"Each particle is at its highest point when they collide" is two conditions in one: both particles must take the *same time* to reach the top, and they must be at the *same place* then. The time condition links U to V, and the horizontal distances then add up to a. After the collision, the vital observation is that both particles are moving horizontally at the top, so the combined particle starts with no vertical velocity at all.

Solution

We model both particles as projectiles moving under gravity alone (no air resistance), in the vertical plane containing AB. Take the direction from A to B as positive horizontally and upwards as positive vertically.



Showing $U^2 = 2V^2$

The particle from A has initial velocity components $\frac{V}{\sqrt{2}}$ horizontally and $\frac{V}{\sqrt{2}}$ vertically. The particle from B has components $\frac{\sqrt{3}}{2}U$ horizontally (towards A) and $\frac{1}{2}U$ vertically.

A projectile is at its highest point when its vertical velocity is zero. With initial vertical speed w this happens at time w/g. Both particles are launched at the same instant, so if they are both at their highest points at the moment of collision, these times must be equal:

$$\frac{V}{\sqrt{2}\,g} = \frac{U}{2g} \quad \Longrightarrow \quad U = \sqrt{2}\,V \quad \Longrightarrow \quad U^2 = 2V^2,$$

as required. Call this common time $T = \dfrac{V}{\sqrt{2}\,g}$.

Ask yourself Shouldn't we also check that the two highest points are at the same height?

Yes: equal times is necessary, but the particles must also meet. A projectile with initial vertical speed w rises to height $w^2/(2g)$. For A's particle this is $\frac{V^2}{2}/2g = \frac{V^2}{4g}$, and for B's it is $\frac{U^2}{4}/2g = \frac{2V^2}{8g} = \frac{V^2}{4g}$. These agree, so the heights match automatically once $U^2 = 2V^2$. (In fact this has to happen: both have zero vertical velocity at the same time T under the same acceleration, so their heights are $\frac{1}{2}gT^2$ for both.) Write

$$h = \frac{V^2}{4g}$$

for the height of the collision.

Finding a

The horizontal positions must also agree at time T. Horizontal velocities are constant, so in time T the particle from A travels $\frac{V}{\sqrt{2}}T$ towards B, and the particle from B travels $\frac{\sqrt{3}}{2}UT$ towards A. They meet, so these distances add up to a:

$$\begin{aligned} a &= \left(\frac{V}{\sqrt{2}} + \frac{\sqrt{3}}{2}\cdot\sqrt{2}\,V\right)\frac{V}{\sqrt{2}\,g} \\ &= \frac{V^2}{g}\left(\frac{1}{2} + \frac{\sqrt{3}}{2}\right) = \frac{V^2}{2g}(1+\sqrt{3}), \end{aligned}$$

which is the required result.

After the collision

At the instant of impact each particle is at its highest point, so each is moving *horizontally*: A's particle with velocity $+\frac{V}{\sqrt{2}}$ and B's with velocity $-\frac{\sqrt{3}}{2}U = -\frac{\sqrt{3}\,V}{\sqrt{2}}$.

Ask yourself Which momentum is conserved, and in which direction?

During the very short impact the only large forces are the equal and opposite forces between the particles; gravity is finite and acts for a negligible time. So total momentum is conserved. Horizontally, with v the velocity of the combined particle of mass $2m$,

$$2m\,v = m\frac{V}{\sqrt{2}} - m\frac{\sqrt{3}\,V}{\sqrt{2}} \quad\Longrightarrow\quad v = -\frac{(\sqrt{3}-1)V}{2\sqrt{2}}.$$

The negative sign says the combined particle moves back towards A, with speed

$\dfrac{(\sqrt{3}-1)V}{2\sqrt{2}}$. Vertically, both particles had zero vertical velocity, so the combined particle starts with zero vertical velocity too.

The combined particle is now a projectile launched horizontally from height $h = \frac{V^2}{4g}$. Its horizontal velocity stays $|v|$. Its vertical speed w on reaching the plane follows from $w^2 = 0 + 2gh$:

$$w^2 = 2g \cdot \frac{V^2}{4g} = \frac{V^2}{2}, \qquad w = \frac{V}{\sqrt{2}}.$$

The angle ϕ that the velocity makes with the horizontal therefore satisfies

$$\begin{aligned}\tan\phi = \frac{w}{|v|} &= \frac{V}{\sqrt{2}} \cdot \frac{2\sqrt{2}}{(\sqrt{3}-1)V} = \frac{2}{\sqrt{3}-1} \\ &= \frac{2(\sqrt{3}+1)}{(\sqrt{3}-1)(\sqrt{3}+1)} = \frac{2(\sqrt{3}+1)}{2} = 1+\sqrt{3},\end{aligned}$$

rationalising the denominator in the second line. This completes the question.

Checks

- *Units.* V^2/g has units $(\mathrm{m\,s^{-1}})^2/(\mathrm{m\,s^{-2}}) = \mathrm{m}$, as a distance should, and $\tan\phi$ is a ratio of speeds, so dimensionless. Note that $\tan\phi$ does not depend on V or g at all, which is a good sign.
- *Sanity of the geometry.* A's particle reaches its top a horizontal distance $\frac{V^2}{2g}$ from A; B's reaches its top $\sqrt{3}\,\frac{V^2}{2g}$ from B. The shallower 30° launch at the larger speed covers more ground, and the two distances add to a.
- *Numerical.* With $V = 7.3\ \mathrm{m\,s^{-1}}$ and $g = 9.8\ \mathrm{m\,s^{-2}}$ we stepped both trajectories forward in time and found them at the same point, $(2.719, 1.359)$ in metres, with both vertical velocities zero, at $t = T$. Stepping the combined particle down to the plane gave $\tan\phi \approx 2.73205$, matching $1+\sqrt{3} = 2.732050\ldots$

Common slips

- Conserving momentum in the vertical direction with non-zero vertical velocities. At the top of each trajectory the vertical velocity is zero, which is exactly what makes the collision so simple.
- Adding the horizontal momenta instead of subtracting: the particles are moving in opposite directions.
- Using $\tan\phi = |v|/w$ (the angle with the vertical) rather than $w/|v|$.

Beyond STEP

Coalescing particles are the simplest example of a perfectly inelastic collision: momentum is conserved but kinetic energy is not (here a good deal is lost, since the particles were moving in opposite directions). At university this becomes the study of collisions in the centre-of-mass frame, where the total momentum is zero and the energy lost is the whole of the kinetic energy relative to the centre of mass. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats momentum and collisions in this way with many excellent problems.

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